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Exploring layered voice analysis: Performance and maze of mechanism

Noé Xiu [a,b], Wenmei Li [a,*], Yuan Wang [a], Béatrice Vaxelaire [b], Rudolph Sock [b], Fabrice Marsac [b], Zhenhua Ling [a]

[a] Interdisciplinary Research Center for Linguistic Sciences – University of Science and Technology of China, Hefei 23000, China
[b] U.R. 1339 Linguistique, Langues et Parole (LiLPa) and Institut de Phonétique de Strasbourg (IPS) - University of Strasbourg, Strasbourg 67000, France

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ABSTRACT

As science and technology continue to advance, sentiment analysis is becoming increasingly important in many scenarios. Especially in areas such as mental health and customer service, there is an increasing need to be able to accurately recognize and analyze human emotions. In this context, Layered Voice Analysis (LVA) technology, which claims to be able to analyze the emotional state of a speaker through speech, has emerged. This technology claims to be useful not only for emotion detection and spoofing detection, but also for helping organizations to enhance user experience in customer service. In this paper, we review the working mechanism of LVA technology and its research progress, and discuss in detail the practical applications of the technology in several fields, including but not limited to the judicial field, the financial field, and the human resource management field. On this basis, we further analyze the challenges faced by LVA in practical applications as well as the prospects. With the continuous progress of technology and optimization of algorithms, speech analysis sentiment tools may show their unique value and wide application potential in more fields in the future.

1. Introduction

Speech analytics is a form of speech data mining beyond mere transcription and voice search to synthesize actionable, instructive information from multiple conversations (Melamed and Gilbert, 2011). Speech analysis technology, as a cross-disciplinary technology integrating acoustics, signal processing, artificial intelligence and other disciplines, has shown strong application value and development potential in many fields such as customer service, medical diagnosis, etc., and has become an important and widely explored direction in forensic science by researchers (Campbell et al., 2009; Eriksson and Lacerda, 2007).

In the field of forensics, the key to speech analysis lies in mining the deeper clues in speech that go beyond words. Human speech itself is an extremely rich treasure trove of information, which can convey a large amount of paralinguistic information in addition to the semantic content of what is said. On the one hand, this information can reflect the physical characteristics of the speaker, such as age, gender, health status and body size, and on the other hand, it also contains emotional states, including positive emotions such as happiness, excitement, thrill, as well as negative emotions such as fear, nervousness, stress, anxiety, and so on (Giddens et al., 2013; Gobl and Chasaide, 2003). In accordance with the appraisal theory of emotion, it is established that human emotional states are responses to an evaluation or assessment of an event or stimulus (Arnold, 1960; Lazarus, 1991; Roseman, 1984). For example, a positive emotional state is elicited when an individual receives an honor or award, while a negative emotional state is elicited when he or she suffers a setback. Additionally, triggering emotions can be an external event, such as a sudden loud noise, or an internal event, such as a physiological change. The core of speech emotion detection lies precisely in extracting emotion-related features from speech for analysis (Sudhakar and Anil, 2015). Some researchers investigated the acoustic features of Chinese speech in different emotional states. Their research results show that the voice characteristics of the three moods—anger, joy, and sadness—are altered to varying degrees in comparison to a calm voice. In a joyful mood, the speaker exhibits an increase in the first formant, F1, which is higher than that observed in angry and sad states. Additionally, a breathing sound is present, the length of speech is shortened, the speed of speech is accelerated, and the fundamental frequency rises. Speech duration is shortest, and speech rate is fastest

* Corresponding author at: Interdisciplinary Research Center for Linguistic Sciences, University of Science and Technology of China- Hefei, No.96, Jinzhai Road, Hefei 230026.
E-mail addresses: xiunoe@ustc.edu.cn (N. Xiu), lwmqt@mail.ustc.edu.cn (W. Li), YuanWang1204@outlook.com (Y. Wang), vaxelair@unistra.fr (B. Vaxelaire), sock@unistra.fr (R. Sock), f.marsac@unistra.fr (F. Marsac), zhling@ustc.edu.cn (Z. Ling).

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when the speaker is in an angry mood. Speakers in an angry, emotional state exhibit the shortest speech duration, fastest speech rate, and the greatest range of vocal tones. In contrast, speakers in a sad emotional state display longer speech duration, slower speech rate, and nasality in pronunciation (Gao et al., 2005). Other researchers have also found that shimmer and the standard deviation of F0 (F0 SD) are indicators of tension and depression in both males and females (Park et al., 2011). Although some studies have revealed potential associations between emotional states and speech acoustic features, it has not yet been established that any one acoustic feature or set of acoustic features can be used consistently and reliably to discriminate between emotions.

Emotional processes can influence cognitive processes, and cognitive processes can modulate or change our emotions; both interactions can be realized by changing mental state variables (Salzman and Fusi, 2010). Thus, the decoding of emotional information that may be embedded in speech provides some basis for our understanding of the connection between speech and internal mental states. However, in many practical application domains, especially in the justice and security domains, our goal is often taken a step further, and we wish to determine the truthfulness of a statement through speech, i.e., to perform deception detection. Deception detection is a far more complex subject than emotion recognition because it involves not only felt emotions, but is also affected by multiple factors such as arousal, attempted control, and cognitive difficulty (Greene et al., 1985). Speech behavior involves the use of multiple sensory modalities, high-level cognitive functions, complex cortical processing, and a large number of motor behaviors and is a manifestation of the output of a high-level integrated nervous system (Hickok et al., 2011; Murphy et al., 1997). Researchers, therefore, believe it is possible to explore more subtle executive operations from speech, such as recognizing deception or truth (Gombos, 2006). A substantial body of literature addresses the relevance and reliability of paralinguistic cues in detecting deception. Anolli and Ciceri analyzed vocal cues and strategies employed by liars. Their findings revealed that deception activates emotional arousal, leading to increased fundamental frequency (F0), greater pausing, and higher word count (Anolli and Ciceri, 1997). Zhang et al. studied the natural speech of players in a social reasoning game and found that when people lie, their voice pauses are longer and more frequent due to heightened cognitive demands. However, lying is not reliably associated with vocal pitch (Z. Zhang et al., 2022). Xiu et al. detected deception through acoustic analysis measuring psychological arousal induced by changes in emotions generated by deception (i.e., fear/stress) and found that the mean values of Voice Onset Time (VOT) and some other acoustic features changed accordingly with disturbances in psychological arousal, and emphasized that acoustic determinations of deception should be relative to the pattern of change in the individual itself (Xiu et al., 2024). However, some studies have also indicated no significant correlation between acoustic characteristics and lying. For instance, in 2011, Kirchhübel and Howard from the Audio Laboratory, Department of Electronics, University of York, UK, investigated changes in speech signals when people lie. In an interview setting, they collected truthful, deceptive, and control speech samples from 10 speakers and analyzed parameters including fundamental frequency, intensity, & vowel formant frequencies. The results revealed that no acoustic feature showed a significant correlation with lying (Kirchhübel and Howard, 2011). Nevertheless, identifying deception through the analysis of paralinguistic characteristics has emerged as a crucial application in forensic speech analysis.

Faced with the complexity of intertwined cognitive and emotional factors in deception detection, the research community has begun to seek a more integrated means of analysis that can decode and analyze multiple speech cues simultaneously. It is against this backdrop that a technology for intelligent speech sentiment analysis exists—Layered Voice Analysis (LVA). LVA technology claims to be based on a set of voice parameters associated with human emotions (these parameters were identified from audio recordings collected across different languages and diverse scenarios, including police interrogations, call centers, and control experiments). These parameters can be used to detect emotional cues in a subject's speech, reflecting various levels of stress, cognitive processes, and emotional responses within their voice. The technology can even identify deceptive intent in “real-life” scenarios. In 2014, the latest version of the core technology (LVA7 CORE) placed greater emphasis on its unique ability to explore and understand different emotional states and personality traits compared to previous versions (Nemesysco, 29 November 2024[1]). Previously, LVA was comprised of two fundamental formulas, each encompassing a distinctive signal-processing algorithm, which was capable of extracting over 120 parameters from each speech segment. The latest version, released in 2014, comprises three basic formulas (consisting of unique signal-processing algorithms) that extract 151 parameters from each speech segment. The aforementioned parameters are capable of recognizing 16 distinct emotional states.

Multiple products have been developed based on LVA technology, such as LVA-i, InTone, QA7, LVA6.5, and InTone.ai (Nemesysco, 29 November 2024[1]). Among these products, LVA-i integrates speech analytics technology with customized questionnaires designed to help organizations conduct pre-employment screening of prospective employees during the recruitment phase and to conduct periodic assessments of their existing workforce. These assessments can be used to evaluate an individual's personality traits and emotional stability (Nemesysco, 29 November 2024[2]). InTone, in contrast, was developed for the call center sector to enable operators to discern the genuine emotional states of customers during service calls or sales processes (Nemesysco, 29 November 2024[3]). Similarly, QA7 is a tool that is used to analyze customer emotions. Given the importance of the customer in business operations, the tool was developed to meet customer needs and focus on customer satisfaction (Nemesysco, 29 November 2024[4]). LVA6.50, conversely, is a specialized investigative tool that claims to be able to detect psychophysiological reactions of suspects and assist in the uncovering of the truth (Nemesysco, 29 November 2024[5]). InTone.ai Risk is a risk assessment system that claims to be able to detect distinctive voice characteristics in insurance claim calls that may indicate an elevated risk of fraud or the withholding of information (Nemesysco, 29 November 2024[6]). In essence, the fundamental tenets of these products are largely analogous. They are all constructed upon the foundation of LVA technology, which is primarily utilized to discern the emotional disposition of the speaker, in addition to psychophysiological responses, and so forth. Presently, LVA technology is already operational in select domains across multiple countries, including China, Japan, and India.

In order to deeply investigate the working mechanism, research progress, application cases, and other related contents of LVA, we searched the databases Google Scholar, Web of Science, Scopus, and PubMed, as well as online public sources using the subject search terms “Layered Voice Analysis” and “LVA”. Included studies had to include tests or applications of layered voice analysis. By the end of 2025, out of the 57 research papers, reviews, commentaries, journals, patents, and other materials searched, 49 met the inclusion criteria, of which 25 were research papers, with a focus on the detailed descriptions in Section 3. Data were independently extracted and cross-checked by two authors.

[1] Nemesysco. 29 November 2024. https://www.nemesysco.com/lva-technology/

[2] Nemesysco. 29 November 2024. https://www.nemesysco.com/recruitment/

[3] Nemesysco. 29 November 2024. https://www.nemesysco.com/intone/

[4] Nemesysco. 29 November 2024. https://www.nemesysco.com/qa7-technology/

[5] Nemesysco. 29 November 2024. https://www.nemesysco.com/security/

[6] Nemesysco. 29 November 2024. https://www.nemesysco.com/insurance/

2. Layered voice analysis

2.1. Working mechanism

To provide an example of an input speech sample, the LVA speech system performs the process of segmentation, which involves the division of continuous speech into a number of discrete speech segments. These segments are characterized by their logical coherence, meaningful content, and absence of background noise. In general, the LVA system selects the initial speech segments or the most suitable speech segments within the recording to establish the baseline for the assessment of an individual's neutral mood. Upon entering the test speech into the LVA system, the LVA extracts the parameters of the test speech and analyzes them in comparison with the speech parameters at the baseline. This analysis generates the basic affective variables, including the mood level and cognitive level, among others. Based on these fundamental variables, additional variables such as arousal and stress can be derived (Mayew and Venkatachalamm, 2012). Furthermore, LVA is capable of processing emotional parameter data through statistical learning algorithms to predict the probability of deception in a sentence.

The latest version of the Layered Voice Analysis core technology (LVA7 CORE) was released in 2014. The voice analysis workflow for this technology is illustrated in Fig. 1. This figure illustrates the speech analysis process of the aforementioned technology. As illustrated in the figure, the LVA system extracts 151 speech parameters from the subjects at baseline and at test time, respectively, and subsequently obtains 16 categories of emotional states through comparative analysis. Depending on the objective of the analysis, LVA also applies specific formulas and algorithms based on these data in order to derive results such as emotional diamonds, lie pressure, and the probability of a sentence having deception.

LVA provides insight into human emotional and cognitive states primarily through voice frequencies, modulations, and patterns (Chaube and Liberman, 2024). As the technical processes of LVA are largely proprietary, two patents related to LVA will be presented in order to facilitate a deeper comprehension of the underlying operational mechanisms. Both patents relate to the detection of a subject's emotional state through speech analysis techniques.

2.1.1. Apparatus and methods for detecting emotions

This patent proposes a method of detecting an individual's emotional state by generating intonation information from the individual's speech specimen and generating an output indication of the individual's emotional state based on the intonation information. In particular, the intonation information herein relates to "Thorn" and "Plateau" as defined in the patent. The "Thorn" and "Plateau" are calculated from n speech samples of 0.5 s duration. "Thorn" means that if the middle sample has the highest or lowest amplitude among three neighboring samples, it is counted as a "Thorn". "Plateau" refers to local flatness, i.e., if the amplitude difference between consecutive samples is less than a given threshold, the sequence is considered to be flat. For n speech samples, the patent obtained the four variables SPT, SPJ, AVJ, and JQ by calculating the number of "Thorn" and "Plateau" and the length of each "Plateau"(Liberman, 2003). Table 1 gives the calculation of these four variables and the corresponding emotional indicator.

In addition, the patent calculates the values of the four parameters SPT, SPJ, AVJ and JQ for the speaker when in a truth/neutral emotional state and when under stress, respectively, and then calculates the deviation of the speaker when under stress relative to when in a truth/neutral emotional state, which in turn gives the following decisions: truthfulness, sarcasm, excitement, confusion or uncertainty, heightened arousal, voice control, lying, exaggeration/inaccuracy. In particular, the patent argues that SPT information is primarily used to determine the excitement level, SPJ information is primarily used to determine the feelings of psychological dissonance. The JQ information is primarily

Table 1
Description of variables.

Variables	calculation method	emotional indicator
SPT	Average number of Thorns per sample	A numeric value describing a relatively high frequency range that patents correlate with the level of emotional stress
SPJ	Average number of Plateau per sample	A numeric value describing a relatively low frequency range. Patents correlate this score with cognitive stress levels
AVJ	Average of the Plateau length	A numeric value describing the average of the relatively low frequency range. Patents correlate this value with the level of thinking
JQ	Standard error of the Plateau length	A numeric value describing the uniformity of the distribution of the relative low frequency range. Patents correlate this score with overall stress levels

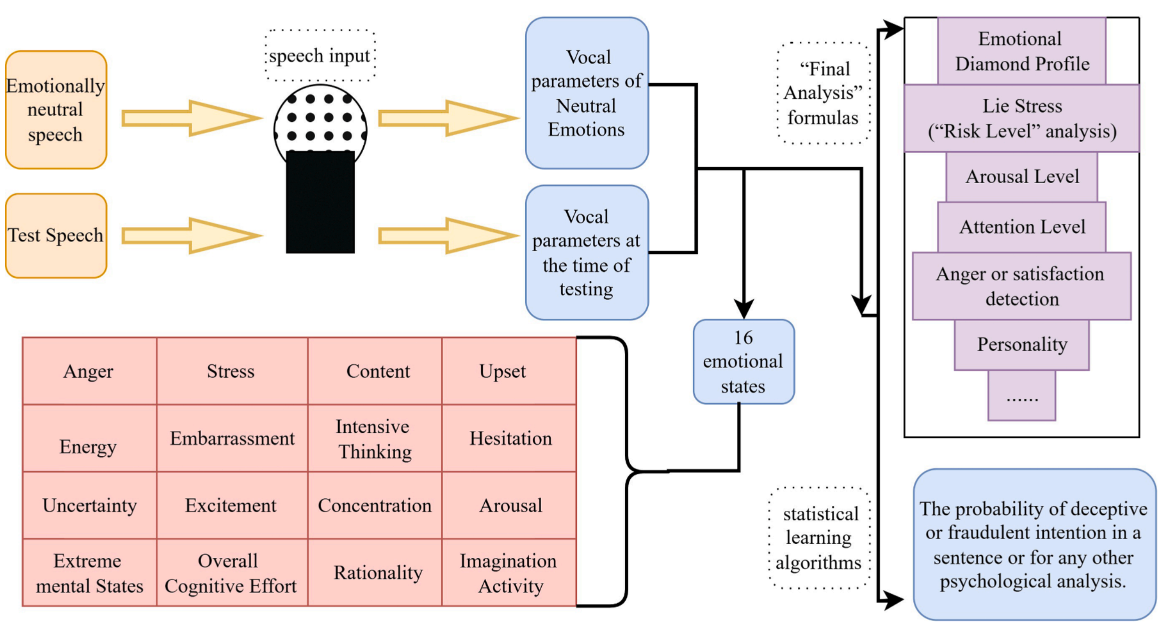

Fig. 1. Flowchart of LVA technology analysis.

related to stress level. AVJ information, on the other hand, has been used to determine the amount of thought invested in spoken words or sentences.

Although LVA technology is constantly being updated, variables such as SPT, SPJ are still the basic variables calculated in LVA. In 2011, Arake and Takeuchi from Japan used the SPT variable to measure pilots' mental tension. Based on the results of their study, they concluded that the emotional stress index SPT can be used to assess the level of mental tension of pilots during flight training (Arake and Takeuchi, 2011).

However, we have no way of knowing whether the current LVA system has been altered relative to this patent's calculation of SPT, SPJ, etc., because the LVA technology is confidential.

2.1.2. Apparatus and methods for detecting emotions in the human voice

The patent is for monitoring a speaker's emotional state, such as attention level, sub-conscious emotional reaction, sub-conscious cognitive activity, and anticipation level, by analyzing his or her voice. The patent uses a specific algorithm to calculate 16 frequency spectra from speech, and based on these frequency spectra, the above mentioned emotional state indicators are calculated (Liberman, 2004). Table 2 gives the calculation of these four emotional states and the corresponding emotional indications.

Based on Table 2, it can be seen that the patent focuses on calculating the selected frequency spectrum in speech and then obtaining an indication of the speaker's general mood based on these frequency spectra. Both patents analyze the speaker's voice to determine the emotional state of the subject. Although the LVA technology is closely related to these two patents, the current LVA system has been significantly updated and the speech features extracted and emotional state calculated have been updated compared to these two patents.

2.1.3. Challenges facing LVA technology

LVA technology claims it can be used to detect both emotions and lies. (Nemesysco, 29 November 2024[1]). While most of the internal processes of LVA technology remain proprietary trade secrets, it is clear that LVA technology faces the same fundamental challenge as other emotion or lie detection technologies: obtaining a reliable ground truth.

Ground truth in emotion detection: Emotion detection aims to identify and understand human emotional states. Ground truth for emotion data annotation can be obtained by having participants classify or rate emotions based on subjective feelings; alternatively, it can be predefined by researchers. Researchers induce target emotions in participants by presenting specific stimuli or designing scenarios, while simultaneously recording participant responses. While establishing ground truth for emotional data may seem straightforward, it is actually a major challenge in emotion research. First, emotions are highly subjective; different individuals may exhibit significant variations in emotional responses to the same stimulus material. This leads to considerable inter-individual differences in ground truth obtained through participant self-reports. Second, emotional expression and experience are strongly influenced by specific contexts. In laboratory settings, it is often difficult to elicit entirely natural emotional responses. In the real world, emotions are complex, variable, and elusive, rarely occurring in isolation but rather as mixtures. These challenges make obtaining authentic and accurate ground truth exceptionally difficult (Constantine and Hajj, 2012). LVA technology claims to be able to detect multiple emotional states. However, obtaining the ground truth for these states, whether in a laboratory or real-world setting, is inherently a challenging issue. Inaccurate ground truth fundamentally casts doubt on the reliability of their technology.

Table 2
Description of mood states.

Emotional states	Calculation method	Emotional indicator
Sub-conscious cognitive activity	Calculated using the sum of the three lowest frequencies	Higher values indicate dense thinking, internal logical conflicts or some cognitive dissonance
Attention level	The most dominant frequency, i.e., the frequency with the largest waveform contribution among the 16 frequencies	It indicates the level of attention or concentration
Sub-conscious emotional reaction	Values calculated with frequencies 11, 12, 13	Higher values indicate stronger emotional responses
Anticipation level	Values calculated with frequencies 14, 15, 16	The higher the value, the higher the degree of anticipation in the subject's voice

Ground truth in lie detection: Lie detection primarily aims to distinguish between truthful statements and false statements. In laboratory settings, ground truth can be obtained through self-reports from participants, where they voluntarily disclose which statements were truthful and which were deceptive after completing the experiment. Ground truth can also be acquired through induced experiments, where researchers design experimental scenarios in advance and explicitly instruct participants to tell the truth or lie under specific conditions (Vrij, 2019). Although this method of obtaining ground truth through induced experiments can yield clear and controllable labels, the lies elicited in such laboratory settings may lack the intense emotions and cognitive load associated with "high-stakes" situations in the real world. In real-world scenarios, ground truth must be determined by verifying the authenticity of statements through conclusive evidence. However, the cost of obtaining such data is high, and large-scale collection is challenging (Curci et al., 2019).

Although the field of lie detection does yield more accurate ground truth compared to emotion detection, it is important to clarify that according to the description of LVA technology, after extracting relevant parameters from speech segments and calculating emotional parameter data, LVA technology can further process this information through statistical learning algorithms to predict whether deceptive or fraudulent intent exists within a sentence (Nemesysco, 29 November 2024[1]). Yet as discussed earlier, while emotion theories continue to evolve, obtaining accurate ground truth in emotion detection remains a persistent challenge. If LVA itself faces skepticism in emotion detection, then deception detection based on statistical learning algorithms applied to emotion data will also be subject to doubt.

2.2. Operating mode of LVA

LVA can operate in two modes, online mode and offline mode (Nemesysco, 29 November 2024[7]) (Kacker and Shukla, 2020), as shown in Fig. 2. The core algorithms of these two modes are the same, only the data acquisition and processing processes are different. Through the combination of these two modes, LVA technology can provide both immediate feedback in real-time situations and deep mining in subsequent analysis, thus providing multi-dimensional speech analysis support for areas such as investigation, psychological research, and human resource management.

2.2.1. Online mode

LVA's online mode is designed for fast and instantaneous analysis of interviews, with a high degree of automation and simplicity, and with low operator requirements. It allows real-time analysis while the speaker is speaking, for example, when a company is hiring a new employee or talking to a customer. The online model is based on real-time calculations of core parameters related to lie stress, global stress, emotional stress, cognitive stress, thinking level, anticipation level, etc. Each parameter is generated by an independent algorithm with speech

[7] Nemesysco. 29 November 2024. https://www.nemesysco.com/lva-6-50-modes-of-operation/

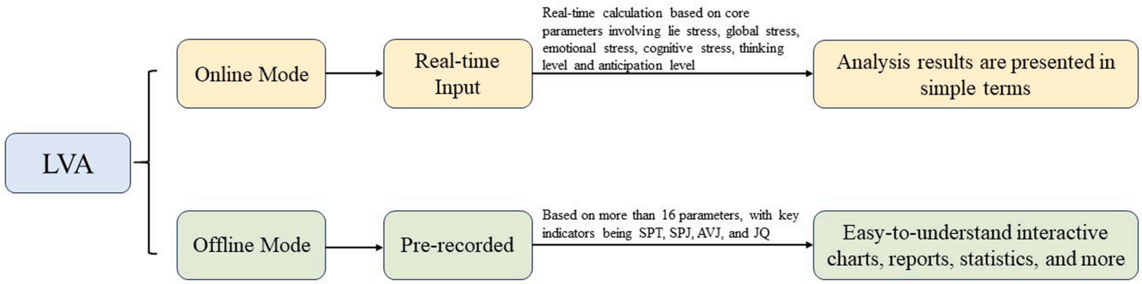

Fig. 2. Operating mode of LVA.

frequency features. The analysis results are presented in concise terms such as truth, high stress, inaccuracy, etc., which can show the emotional and honesty state of the participant sentence by sentence, clearly reflecting his/her psychological dynamics.

2.2.2. Offline mode

The offline mode is mainly used for systematic analysis of pre-recorded audio material, and supports a wide range of audio source formats, such as recordings, videos, and broadcasts. The core of the Offline Mode is Rich Psychological Analysis (RPA). This mode allows human participation in interpretation, analyzes the relationship between zones, and provides easy-to-understand interactive charts, reports, statistical information, etc., which facilitates in-depth analysis of the subject's mood swings, psychological conflicts, and behavioral motivations. Unlike the online mode, the offline mode requires the operator to undergo certain professional training to fully understand the meaning of the analysis results and psychological indicators. The offline model is capable of recognizing deception patterns, such as the overall deception pattern and the aggressive lie pattern. The analysis of the offline mode is based on more than 16 parameters, and the algorithm calculates the raw values to form a report, of which the key indicators are SPT, SPJ, AVJ, and JQ.

3. Current status of LVA research

Since the LVA technology was proposed, several countries around the world have paid attention to and researched it. For example, China, the USA, Sweden, India, Japan, and many other countries have conducted in-depth studies on the effectiveness of this technique. The following sections will respectively introduce the current state of LVA research in the fields of emotion detection and lie detection, with Table 3 attached.

3.1. Emotion detection based on LVA

As early as 2008, Japanese medical researchers Nemoto et al. began exploring the use of LVA to detect psychological stress. The study recruited 106 healthy participants, randomly assigning them to an experimental group and a control group. Before task execution, all subjects underwent measurements of the State-Trait Anxiety Inventory (STAI), blood pressure, and salivary amylase levels, while their voices were recorded. Subsequently, the experimental group performed an anagram task, while the control group read aloud task instructions. Results revealed significant correlations between speech parameters and trait anxiety and blood pressure before task execution. Furthermore, speech parameter variations during anagram solving were significantly greater in the experimental group than in the control group. The authors concluded that these findings suggest LVA technology holds potential for detecting psychological stress (Nemoto et al., 2008).

In the same year, Hada and Takeuchi from the Japan Air Self-Defense Force and the Aviation Medical Research Institute first validated the assessment value of LVA for pilot mental stress in a flight simulator scenario. Using 14 male JASDF members as subjects, they simulated a high-stress crosswind landing scenario in a flight simulator while simultaneously measuring participants' heart rates and voice data. LVA extracted nine voice parameters related to stress, including Brain Power and SPT (directly correlated with emotional stress, where higher values indicate greater tension). Their findings revealed that when constructing a multiple regression model with heart rate as the dependent variable and the aforementioned nine LVA voice parameters as independent variables, the model's predicted heart rate trends aligned with actual heart rate changes. The study concluded that voice analysis technology can estimate pilot heart rate variations through LVA parameters, offering potential as a tool for assessing mental stress in aviation scenarios. However, interpreting the study's findings reveals that while Brain Power showed a positive correlation with heart rate, the remaining eight stress-related parameters exhibited negative correlations. For instance, the partial correlation coefficient of SPT is −0.130, which indicates that when other variables remain constant, an increase in SPT leads to a decrease in heart rate and a reduction in tension, contradicting the definition of using SPT itself (Hada and Takeuchi, 2008). Additionally, Takeuchi and Arake from the same laboratory conducted another study in 2010 involving seven healthy male participants (mean age 42.14 years, SD=10.22) as subjects. Participants read aloud questions from the Profile of Mood States (POMS) displayed on a screen and responded verbally while their speech was recorded. LVA analysis was then applied to the speech data collected during responses to each question, examining correlations between LVA parameters and POMS subscales. Their findings revealed significant positive correlations between POMS T-A (Tension-Anxiety) and A-H (Anger-Hostility) scores and LVA Lie Stress. Additionally, the POMS F subscale (Fatigue) showed a significant positive correlation with the LVA “JQ” parameter. However, prior research by other investigators concluded that the POMS F subscale negatively correlated with the JQ parameter. The two studies yielded opposite results, leading Arake and Takeuchi to question the measurement stability of LVA. They concluded that LVA cannot effectively detect negative psychological emotions from the speech of healthy adults (Arake and Takeuchi, 2010). However, in 2011, Arake and Takeuchi conducted a study with nine male flight trainees, collecting their voice samples at early, middle, and late stages of the pilot training process. They correlated the SPT (emotional stress index) related to mental tension in LVA with the instructors' assessments of the trainees' skill performance. The results showed a significant negative correlation between SPT and flight skill performance in the early and middle stages. They concluded that SPT could serve as an effective indicator for assessing mental tension during pilot training (Arake and Takeuchi, 2011). The studies described above suggest that early in the development of LVA, Japanese researchers were already concerned about LVA's ability to detect mood. However, for some of the mood indices in LVA, different studies have yielded opposite results, which has led some researchers to question the stability of LVA indicators.

In 2012, U.S. financial researchers Mayew and Venkatachalam used LVA to analyze conference call audio files to examine the emotional state of managers during earnings calls. They found that the positive and

Table 3
Status of research on LVA.

Author **Emotion detection**	Source of speech data	Research Topics
Nemoto, Tachikawa, Tanimukai et al. (2008)	Voice recordings were taken from 106 healthy subjects divided into experimental and control groups while performing tasks.	Using LVA to detecte psychological stress
Hada and Takeuchi (2008)	Voice data of 14 male Air Self-Defense Force members in high-stress flight scenarios	Using LVA to evaluate mental stress during simulated landing
Arake and Takeuchi (2010)	Oral responses of 7 subjects reading the POMS scale items	Correlation between LVA and Profile of Mood States (POMS)
Arake and Takeuchi (2011)	Pilot's voice during training	Using LVA to assess pilot stress levels
Mayew and Venkatachalam (2012)	Conference call audio files	Using LVA to detect emotional states exhibited by managers during earnings calls
Hobson, Mayew and Venkatachalam (2012)	Telephone conference audio recordings and experimental recordings of 59 university students' speech	The Predictive Role of CEO Cognitive Dissonance Voice Markers in Financial Misstatements During Earnings Conference Calls
Price, Seiler and Shen (2017)	Sample audio from quarterly earnings conference call	Using LVA to analyze the emotional content implied by managers' voices in quarterly earnings conference calls as a way to study investor reactions to emotional information.
Műhelyi, Kis and Takács (2022)	Participants assigned to different identities were interrogated after completing the interactive role-playing game "Mafia" (also known as Werewolf)	Using LVA to detect participants' cognitive load and emotional load
Pusker, Berényi,T. Kárász and Takács (2025)	16 undergraduate students simulated the voice of a psychology master's program application interview	Using LVA to detect the emotional patterns of subjects in four types of questions (irrelevant questions, professional questions, professional challenge questions, ten-year plan questions)
Majumdar and Rupaali (2023)	Voice samples of suspects	Differences between male and female criminal suspects in levels of emotional reactivity were compared using the LVA comparison
Tani, Takao, Noro et al. (2024)	Voice calls between call center operators and customers	Organisational intervention with call centre operators using LVA to analyse its impact on mental health
Tani, Takao, Noro et al. (2025)	Daily call recordings of 62 employees in the call center	Using a voice and emotion analysis system based on hierarchical speech analysis, speech at four time points—baseline, 4 weeks, 8 weeks, and 12 weeks—is analyzed to detect depressive symptoms
Zhang, Zhang, Wang et al. (2024)	Live streamer's verbal responses in live streaming scenarios	Using LVA to quantify emotional support at the vocal level in streamers' verbal responses
Hsu, Lin, and Chung (2025)	2014–2021 Significant Information Press Conference Management Speech for Listed Companies in Taiwan Stock Exchange and Taipei Exchange, China	Measuring voice uncertainty in the Q&A session through LVA software

Table 3 (*continued*)

Author **Emotion detection**	Source of speech data	Research Topics
Deception detection		
Sommers, Brown, Senter et al. (2002)	Collect participants' corresponding speech through different scenario settings.	Validating the effectiveness of LVA in laboratory settings
Damphousse, Pointon, Upchurch et al. (2007)	Voice samples of suspected drug users in prison	The validity and reliability of LVA were tested in a real environment
Damphousse (2008)	Statements from 319 arrestees in Oklahoma County Jail regarding recent drug use	Compare the urine test results to analyze the effectiveness of LVA
Harnsberger, Hollien Martin et al. (2009)	Speech samples from subjects in a mock crime experiment	Evaluation of LVA performance using material recorded while highly controlled deception and stress levels were systematically varied
Horvath, McCloughan, Weatherman et al. (2013)	Audio data collected at Michigan State Police site	Deception detection using LVA and comparing recognition accuracy between LVA operators and interviewers
Chao, Wang, Liu et al. (2020)	Subjects' voice responses to sensitive questions	Explore the performance of LVA in different situations and analyze the data to find out the factors affecting the recognition rate of the system.
Maniar, Rathod, Kumar et al. (2022)	Fraud calls voice samples	Deception Detection of 12 Fraud Calls Using LVA
Hussain, Srivastava, and Gupta (2024)	Speech samples from subjects in a mock crime experiment	Deception detection using LVA and comparison with two other lie detection tools
K S, Rastogi, and Ramesh (2024)	15 true confessions and 15 false confessions	Using LVA to conduct offline and online analysis on 30 confessions, respectively, and detecting parameters related to deception
Rastogi and Ramesh (2024)	30 voice samples of suicide statements sourced from the internet and laboratory databases	Using LVA to Explore Emotional Patterns in Suicide Statements and the Rate of False Declarations
Marathe, Madhusudan, Kaashika et al. (2025)	The natural speech of the artist, the artist's imitation of the former Indian Prime Minister's speech, and the original speech of the former Prime Minister obtained from the internet	Using LVA to analyze the matching degree of five parameters, including emotional stress and cognitive stress, in three types of speech samples

Note:The same author used identical colours for annotation.

negative emotions managers display have an impact on a company's financial outlook. Analysts do not take this information into account when forecasting short-term earnings. However, when changing their stock recommendations, analysts incorporate positive but not negative influences (Mayew and Venkatachalamm, 2012). However, the approach of Mayew and Venkatachalam to directly analyze emotions in speech using LVA technology has been questioned by some researchers (Lacerda, 2012). In the same year, Mayew and Venkatachalam, along with Hobson from the Department of Accountancy at the University of Illinois at Urbana-Champaign, conducted another study. In their research on whether vocal markers of cognitive dissonance are useful for detecting financial misreporting, they used vocal dissonance markers generated by LVA technology. The study first evaluated the validity of the dissonance markers generated by LVA in a laboratory setting. Since "feelings of cognitive dissonance" cannot be directly measured, the researchers used four indirect indicators as the "true manifestations of cognitive dissonance." They collected four cognitive dissonance proxy variables and voice samples from 59 participants through experiments.

Then, they used LVA to generate "vocal indicators of cognitive dissonance" and analyzed the correlation between these vocal indicators and the four cognitive dissonance proxy variables. The results showed a significant positive correlation between the "vocal indicators of cognitive dissonance" and the cognitive dissonance proxy variables, but the correlation coefficients did not exceed 0.44. The researchers believed that univariate correlations might be influenced by "individual differences" such as age and gender. Therefore, they constructed four multivariate regression analysis models, using the four cognitive dissonance proxy variables as dependent variables and the early "vocal indicators of cognitive dissonance" as independent variables, while controlling for other factors such as age and gender. For each model, the coefficient of the early "vocal indicators of cognitive dissonance" was significantly greater than zero. However, the goodness-of-fit R^2 for each regression model was relatively small, with the largest R^2 being 0.552. The researchers concluded that these experimental studies had thoroughly demonstrated that LVA indicators can capture "cognitive dissonance." Consequently, Hobson et al. used the vocal indicators of cognitive dissonance generated by LVA to assist in detecting misreported financial statements. The study found that the diagnostic accuracy was only 11% higher than random chance (Hobson et al., 2012). However, another scholar has criticized Hobson et al.'s research. While acknowledging the innovative contributions of Hobson, Mayew, and Venkatachalam, the critic raised concerns across three dimensions: "the uniqueness of theoretical explanations," "sample representativeness," and "the boundaries of technological application." The critic directly points out the shortcomings of LVA in financial scenarios. The most fundamental critique centers on Hobson et al.'s assumption that "LVA indicators = cognitive dissonance." The critic argues that these indicators may confound other psychological states, leading to biased interpretations of conclusions (Li, 2012).

In 2017, Price et al. collected audio recordings and corresponding written transcripts of equity real estate investment trust conference calls from Bloomberg for a sample of 309 firm quarterly observations from 2010 through 2011. They similarly used LVA to analyze the emotional content implied by executives' voices during quarterly earnings conference calls, and examined investors' reactions to emotional information through LVAi 7.7-generated emotional activity level variables and overall cognitive activity level variables. They found that managerial emotion is positively correlated with initial investor reactions, and the results are statistically and economically significant. Moreover, the strong investor response to REIT managers' emotional signals seems plausible, suggesting that emotion-related signals contain plausible, value-relevant information. However, the author also found limited evidence suggesting a partial reversal in subsequent trading windows, which the author believes may indicate that investors might be second-guessing themselves or worrying about overreacting (Price et al., 2017).

In 2022, researchers Műhelyi and Takács from Károli Gáspár University of the Reformed Church in Hungary and György from ANIMA Polygraph Ltd. explored the cognitive load and emotional load behind deceptive behavior through instrumental means. The researchers examined the cognitively relevant and emotionally relevant variables given by the LVA speech analysis software and hypothesized that these variables together constitute the cognitive and emotional loads associated with the probability of lying. Specifically, through the definition of the variables by LVA itself, the LVA-generated thinking level, imagination, and cognitive level were selected as the observed variables of cognitive load, and the LVA-generated concentration, confusion, and reaction expectancy were selected as the observed variables of emotional load. The correlations between the cognitive-related and emotional-related variables given by LVA and the latent variables (corresponding to cognitive load and emotional load) were also obtained using the Confirmatory Factor Analysis (CFA) method. The results of the study showed that there were significant correlations between the elements of the model. The researchers concluded that this indicates that cognitive and emotion-related changes in speech can be instrumentally measured and that the two core influences behind lying are cognitive load and emotional load (Műhelyi et al., 2022). Three years later, Takács, along with Pusker et al. from the same school, used LVA software to detect emotional patterns in simulated interviews. The team simulated interviews with 16 full-time psychology undergraduates, ages 20–45, for the Master of Arts in Psychology program, dividing the questions into four categories: irrelevant questions, professional knowledge questions, professional challenges and resources questions, and ten-year plan questions. The interview process was recorded by audio-recording devices and then analyzed with the help of LVA for 15 voice parameters, and finally statistically analyzed using IBM SPSS 28.0 and JAMOVI software, with a total of 3120 valid voice segments collected and analyzed. The results show that the means of the 9 LVA parameters (Aggression, Atmosphere, Brainpower, Content, Energy, Excited, Hesitation, Stress, Upset) differ significantly across different problem categories. Among them, the difference in the Brainpower parameter was both statistically significant and professionally informative ($\chi^2(3)=44.8$, $p < 0.001$). Mean values of the Atmosphere parameter are in the positive range, which, in combination with the data on low hesitancy, low aggression, low upset, and low stress, indicates that the overall atmosphere of the interview was relaxed. It is worth noting that the overall values of the stress parameter were low. No significant stress was detected. The mean stress value was higher for the ten-year planning question than for the specialized knowledge question. This is contrary to the initial hypothesis that specialized questions trigger higher stress. The researchers believe this may be related to the fact that the stress reference scale provided by LVA technology is based on a standard of more extreme scenarios (e.g., life-threatening). There were also no actual college consequences for this interview. The study concluded that different interview question categories elicited unique emotional patterns that could be effectively detected by the LVA software. However, the study has limitations such as small sample size and underrepresentation of gender, and the specialized interpretation of some of the LVA parameters needs to be further validated with more scenarios (Pusker et al., 2025).

In 2023, Majumdar and Rupaali of the Forensic Psychology Division, India, used a sample of 60 criminal suspects (30 males and 30 females) aged 18–40 years to compare the differences between males and females in four types of emotional response levels (emotional level, cognitive level, stress level, and thinking level) using the offline model of the LVA. The results of the study showed that females scored significantly higher than males only on the emotional level, while males scored significantly higher than females on the other three levels. The authors suggest that the specific factors contributing to these differences are complex and multifaceted, including socialization processes, differences in emotional information processing, and hormonal and neurobiological factors. In addition, cultural factors may play a role in shaping emotional expression and experience (Majumdar and Rupaali, 2023).

In 2024, Tani et al. examined the effects of an organizational intervention using LVA on the psychological well-being of call center operators. Participants were randomly assigned to three groups: an organizational intervention using LVA, an intervention of one-on-one meeting, or active control, and psychological status was monitored through a self-assessment of the CES-D Depression Questionnaire at four time points. Results suggested that even though busy call center operators are unable to participate in the LVA intervention program, mental health interventions can be effective if they participate more than 60% of the time. Participatory organizational interventions using LVA have the potential to reduce the risk of depression among call center operators and are more effective in preventing depressive symptoms early on (Tani et al., 2024). Subsequently, Tani et al. conducted a prospective 12-week cohort study of 62 employees of a call center in Kumamoto Prefecture, Japan, for objective early detection of depressive symptoms using a voice and emotional analysis system based on LVA. Depressive symptoms were assessed by the Center for Epidemiologic Studies Depression Scale (CES-D) at four time points (baseline, 4 weeks, 8 weeks, and 12 weeks), while employees' daily work calls were automatically recorded, and 22 acoustic parameters were extracted with the help of the Emotional Signature Analysis System (ESAS). Subsequently, 11 key voice parameters were screened by the recursive feature elimination method, and the probability scores were calculated by logistic regression to construct a comprehensive predictor, "synthetic variable from the probability score (SVPS)", and ROC curves, pseudo-external validation of time segmentation, and generalized estimation equations were used to validate the performance and robustness of the model. The results of the study showed that the composite predictor (SVPS) was significantly and moderately positively correlated with the CES-D score ($r = 0.484$, $p < 0.001$); the AUC of the predictive model was 0.783 (95% CI: 0.691–0.875) and the ratio of employees with SVPS$\geq$0.334 showing depressive symptoms after one month was 7.78 (95% CI: 3.27–18.5), and the sensitivity analysis further confirmed that the model has certain robustness and generalizability. Based on these results, the study concluded that voice sentiment analysis can be used as an objective tool for the early detection of depressive symptoms in the workplace, and that a one-month warning can help occupational health personnel to carry out proactive interventions (Tani et al., 2025).

In 2024, Zhang et al. from China used LVA to quantify the "voice-based emotional support" in streamers' oral responses while studying how these responses convey informational support and emotional support, as well as the impact of these two types of support on user experience. Considering that the emotional delivery of streamers' oral response in live broadcasting scenarios relies on non-textual acoustic features, and that traditional textual analysis is unable to capture this kind of information, the study chooses the LVA technique to realize the deep mining of emotional nuances in the audio, which provides data support for the subsequent empirical analyses of the voice emotion score as a quantitative index. Through the ordered Logit model analysis, it is found that the information support and voice-based emotional support embedded in streamers' responses have a positive effect on the user's live broadcast experience, while the content-level emotional support does not show a significant effect. This result suggests that streamers should adopt the strategy of "delivering more informative content with emotional voice" to enhance user experience. This study highlights the promising application of LVA technology in capturing non-textual emotional delivery paths (M. Zhang et al., 2024).

In 2025, Hsua et al. examined major information press conferences held by companies listed on the Taiwan Stock Exchange and Taipei Exchange in China from 2014 to 2021, focusing on investor reactions to managerial vocal cues in Chinese disclosure contexts. By quantifying management's linguistic tone during statements (measured via positive/negative word proportions using the Chinese version of the LIWC2015 lexicon) and vocal uncertainty during Q&A sessions (measured using LVA software), the study yielded several key findings. When conferences included Q&A segments, both management's linguistic tone and vocal uncertainty provided predictive information regarding the realization of future long-term profits. Analysts incorporate information related to management's vocal uncertainty when adjusting expectations for long-term stock recommendations, but do not consider information related to linguistic tone (Hsua et al., 2025).

Through the aforementioned literature, we can see that only a portion of researchers have investigated the ability of certain LVA emotional variables to reflect genuine emotions, while many researchers tend to directly use the indicators provided by LVA technology to represent corresponding emotions.

3.2. Lie detection based on LVA

As early as 2002, Professor Sommers of psychology in Arts & Sciences at Washington University researched the lie-detection capabilities of LVA technology, funded by the U.S. Department of Defense Polygraph Institute (DODPI). The study designed multiple experiments to validate the effectiveness of LVA technology, but the results consistently showed its accuracy did not exceed random levels (Everding, 2004; Sommers et al., 2002).

In 2007, Professor Kelly Damphousse et al. tested the validity and reliability of the LVA in a real-world setting. They randomly selected several arrestees in a county jail, asked them about their recent drug use, and used the LVA to analyze their speech as they responded to determine whether they were lying. The LVA judgments were then compared with the results of urine tests to assess whether the LVA could detect deception. The results showed that the validity of the LVA was poor, with an average sensitivity for detection of 21%. In terms of specificity, the results were similar for LVA experts and novices. The authors concluded that the method was not effective in identifying those who were deceptive about recent drug use (Damphousse et al., 2007). Subsequently, a field study led by Dr. Kelly R. Damphousse and funded by the National Institute of Justice (NIJ) was conducted on 319 detainees at the Oklahoma County Jail comprehensively validated the effectiveness of LVA. This study focused on detecting lies related to recent drug use by comparing urine test results with VSA analysis outcomes. Regarding core detection metrics, LVA demonstrated outstanding specificity (the proportion of truthful individuals correctly identified), correctly classifying 95% of truthful subjects as "non-deceptive" across five drug types. However, LVA's performance in the more critical sensitivity metric (the proportion of liars correctly identified) remains inadequate. It correctly identified only 21% of drug-use-related lies—while higher than CVSA's 8%, it falls far short of the precision required in real-world law enforcement scenarios. The average sensitivity of 15% across both software programs further highlights LVA's limitations in detecting deceptive signals. Combining sensitivity and specificity calculations, LVA's overall accuracy in detecting drug-related lies is approximately 50%, placing it alongside CVSA at a level comparable to flipping a coin. This fails to meet law enforcement agencies' core requirements for lie detection technology. Nevertheless, like other VSA software, LVA demonstrates some value in "deterring deception." Comparing data from the Arrestee Drug Abuse Monitoring (ADAM) study, which did not employ VSA technology, revealed that when subjects were aware of VSA tools like LVA being used during interviews and informed that lies could be detected, the rate of drug-related false statements dropped to just 14%. This is significantly lower than the 40% false statement rate observed in the ADAM study. This "bogus pipeline effect" indicates that the presence of LVA imposes psychological constraints on respondents, reducing deceptive statements (Damphousse, 2008).

In 2009, Dr. Harnsberger et al. in the United States evaluated the performance of the LVA using material recorded in a highly controlled deception situation with systematically adjusted pressure levels. The results showed that across all conditions, material types (e.g., pressure vs. non-pressure, real vs. deception), and examiners (a pair of scientists trained and certified by the manufacturer and two experienced LVA instructors provided by the company), the "true positive" (or hit) rate for all examiners averaged close to random levels (42–56 percent). Most importantly, the false positive rate was very high, ranging from 40% to 65%. The authors concluded that, as confirmed by the sensitivity statistics, the LVA system performs at a comparable randomized level in detecting truthfulness, deception, and the presence or absence of high and low voice pressure states (Harnsberger et al., 2009).

In 2013, Horvath et al. compared the accuracy of truthfulness and deception judgments of 31 truth tellers and 43 false tellers made by three interviewers and two LVA operators based on audio data (pre-test interviews for polygraph examinations) collected from the MSP (Michigan State Police). The results showed that the average accuracy of LVA operators' judgments of truth-tellers was 48% and the average accuracy of judgments of deceivers was 25%, whereas the average accuracy of interviewers' judgments of truth-tellers was 68% and the average accuracy of judgments of deceivers was 71%. The authors concluded that it was clear from the results that the judgment accuracy of the LVA operators was low and much less accurate than the judgment accuracy of the interviewers who only interviewed the subjects (Horvath et al., 2013).

In 2020, Chao et al. from the Chinese Criminal Police Academy asked questions to 20 research subjects in two scenarios: telling the truth about the experiment (arranging unfamiliar teachers to simulate higher-ups to come to the school to do a questionnaire survey) and not telling the truth about the experiment (the students answered the questions randomly according to their own ideas and were allowed to lie when answering the questions). After the test was completed, the students were talked to separately and asked about the correct answers to the questions and the psychological situation at the time of the test. The voice samples recorded during the experiment were analyzed through the LVA voice system. Their results showed that for subjects who were not told the truth about the experiment, the average recognition rate of LVA was 91.75%, which was much higher than those who knew the truth about the experiment (CHAO et al., 2020).

In 2022, Maniar et al. from the Central Forensic Science Laboratory in India analyzed 12 samples of fraudulent phone calls for spoofing detection using the LVA technique. It was found that LVA identified eight fraudulent calls as having a high risk of speech, and four fraudulent calls for which no conclusion could be drawn. Although the percentage of high risk identified by LVA was 2/3, the authors still concluded that the LVA technique is the most effective tool for analyzing human emotions using voice parameters (Maniar et al., 2022).

In 2024, Hussain et al. designed a mock crime experiment and administered polygraph examinations to three suspects using three polygraph instruments: the LVA, the Suspect Detection System (SDS), and the polygraph. The experiment was designed to have only one perpetrator, but both the LVA and the polygraph resulted in two subjects being convicted as suspects (Hussain et al., 2024). The authors concluded that the major limitation of the study was that the sample size was too small, which was a significant limitation on the inferential results.

In the same year, K.S., Rastogi, and Ramesh selected 30 participants aged 20–30 with no physical, psychological, intellectual, or speech-related issues from the Central Forensic Science Laboratory in Kolkata. They were randomly divided into two groups of 15 individuals each. Group 1 made a threat call to any trusted person, followed by a confession based on the actual behavior. Group 2 was required to memorize a scripted story and provide a false confession strictly adhering to the story's content. The interview process for both groups was recorded. Subsequently, LVA instruments were used to conduct offline and online analyses on 30 confessions, with data processed statistically using both qualitative and quantitative methods. Results revealed that in online analysis, false confessions exhibited significantly higher mean values than truthful confessions across deception-related parameters, including High Risk, Max Lie Stress, Global Honesty Rate, High-Risk Rate, Deceptive Potential, and Deep Falses. Additionally, false confessions showed a strong positive correlation with high-risk rate ($r = 0.617$, $p = 0.000$). In offline analysis, false confessions exhibited higher mean values across deception-related parameters, including High-Risk Samples, Stressed Samples, Inaccuracy Sample, Highly Suspected Segments, and Average Risk Probability. False confessions also demonstrated a weak positive correlation with high-risk samples ($r = 0.400$, $p = 0.028$). Notably, offline analysis demonstrated higher sensitivity, with the intergroup difference in the “high-risk samples” parameter far exceeding that of the “high-risk percentage” parameter in online analysis. The study confirms that LVA can distinguish truthful from false confessions through a series of parameters. Combining offline and online analysis enhances deception detection reliability, providing technical support for preventing miscarriages of justice (KS et al., 2024). Additionally, Rastogi and Ramesh conducted another study analyzing 30 voice samples of suicide declarations sourced from online and laboratory databases using LVA (Layered Voice Analysis) technology. They aimed to explore emotional patterns in suicide declarations and their false declaration rates. Results revealed that the core emotions of suicide declarants were Stressed (mean 17.13) and High-Tension (mean 15.77), with other emotional parameters such as Truth (mean 14.23) and Inaccuracy (mean 12.77) also being significant. The mean values of indicators associated with false statements (e.g., High Risk, Voice Manipulation) were extremely low (0.70 and 0.33, respectively), and no emotional expressions related to low risk or stress relief were observed. This indicates that most of these suicide declarants genuinely expressed their intentions, with an extremely low false statement rate. Except for Extreme Stress ($P = 0.081$), statistically significant differences were observed in the mean values of other emotional parameters, including Stress, Inaccuracy, and High Tension ($P < 0.05$). The study concludes these statements cannot be interpreted as false suicidal statements, highlighting LVA's forensic application in protecting individuals from false accusations by those with delusional suicidal thoughts or attempts, thereby reducing misjudgment probabilities. However, given the limited available information, the actual ground truth of these suicidal statement samples remains unknown, making it difficult to assess LVA test validity (Rastogi and Ramesh, 2024).

In 2025, Marathe et al. focused on the interdisciplinary integration of forensic psychology and audio forensics, aiming to explore the combination of various forensic psychological techniques and audio analysis to provide evidence for the deceptive identification of imitated voices. The study selected one imitation artist as the research subject and collected three types of voice samples, including natural speech, imitated speech of a former Indian Prime Minister, and the original speech of the former Prime Minister obtained from the internet. In terms of analytical methods, parameters such as Jitter and Shimmer were measured using PRAAT software to conduct phonetic and spectral comparisons. Then, an LVA test was performed on the artist's voice to analyze the matching degree of five parameters, including emotional stress and cognitive stress. Subsequently, relevant questions were designed around whether the artist imitated celebrity voices and whether they profited from such imitations, and a polygraph test was conducted. Finally, a BEOS test was administered, where probes related to imitation behavior (such as " Mala Lahanpanapasun mimicry karayla aawadte (I have enjoyed imitating since I was young)") were played to the artist to detect whether related event memories existed in their brain. The results of the study showed that audio analysis revealed a high similarity in phonetic characteristics between the artist's natural speech and imitated speech. In the LVA test, the artist's natural speech and imitated speech showed a complete match across five emotional parameters, both indicating clear signs of deception, while no graphical match was found with the original speech of the former Prime Minister. The polygraph test, through the design of relevant and control questions, detected significant deceptive signals in responses to questions about whether the artist imitated specific celebrities and whether they profited from such imitations. The BEOS test revealed that the artist exhibited an "experiential knowledge" response to probes related to imitation behavior, confirming the existence of memory traces of imitation behavior in their brain. The study ultimately concluded that audio forensics and forensic psychology techniques have complementary applications. At the same time, the study also pointed out limitations, such as the small sample size, which limits the representativeness of the conclusions. Future research should expand the sample size and further validate the findings with voice samples from real cases (Marathe et al., 2025).

Based on the available LVA research, it is evident that researchers have given mixed reviews of its effectiveness. In particular, when LVA is used for polygraph examinations, the results show a clear divergence. On the one hand, several studies have indicated that the accuracy of LVA in detecting lies is not high or even lower than expected. Researchers have suggested that this may be due to a variety of factors, such as operator judgment and limitations of the technology itself. On the other hand, there are some studies that suggest that LVA is more accurate in detecting lies. Therefore, further research and investigation are urgently needed to test the validity and reliability of LVA. This should include not only an in-depth analysis of the existing data, but also more diverse experimental designs to test its applicability in different contexts. In addition, future studies could consider introducing more samples to ensure the generalizability and reliability of the results. Through systematic research, we can better understand the potential of the LVA technique and its limitations in practical applications.

4. Fields of application of LVA

LVA is now employed by a multitude of commercial entities. These applications encompass the justice sector, service assurance in call centers, the HR hiring process, market research, and even medical assessment protocols for a variety of needs. The utilization of LVA in these domains provides a novel avenue for accessing a vast repository of information, facilitating the measurement and analysis of diverse emotional states (Nemesysco, 29 November 2024[1]).

4.1. Judicial field

The use of LVA in the judicial field has been documented in a number of countries. For example, in China, LVA voice systems are employed as an investigative tool in the judicial context (JRVOICE, 2 April 2019[8]). In India, the judiciary employs LVA to administer lie detector tests to suspects. The following is a real-world example: On February 4, 2023, an Indian news report stated that a minister was murdered in broad daylight while attending a political party event. Shot at close range by the suspect, he was airlifted to the hospital in critical condition and passed away that evening. Regarding the suspect in this murder case, a team from the Central Forensic Science Laboratory in New Delhi administered a lie detection test using LVA (OMMCOM NEWS, 4 February 2023[9]).

4.2. Market service field

LVA is also widely used in market research; some companies use LVA to analyze the emotions of their customers in order to understand their preferences and provide better services. For example, on October 6, 2020, a Japanese technology company announced the launch of a service that analyzes emotions based on voice. Beyond analyzing customer sentiment and leveraging it for sales activities, it can also identify stress levels from employee voices (NIKKEI, 6 October 2020[10]). Additionally, according to information on the official website of the company that owns LVA technology, a Japanese call center service provider is utilizing this technology to optimize customer experience and sales performance across its three Japanese call centers. The provider has integrated QA7 technology and layered voice analysis technology into its emotion detection and analysis services—supported by LVA's parent company—enabling real-time feedback on callers' emotional states. According to company officials, these technologies significantly enhance the optimization of outbound marketing and sales activities while enabling precise identification of upselling opportunities during incoming calls (Customer Service Manager, 12 September 2019[11]).

4.3. Financial field

In finance, LVA is used to identify the emotional states of managers as a way to help analysts predict short-term returns (Mayew and Venkatachalamm, 2012). In addition, LVA claims to help insurance companies deal with false insurance claims, identify the risk of fraudulent insurance policies, and reduce insurance company losses. For example, the July 3, 2020 edition of Asia Insurance Review reported that a Chinese insurance company plans to expand the use of voice analytics and sentiment detection solutions over the next two to three years to conduct risk assessment and fraud detection across more insurance types and business units (ASIA INSURANCE REVIEW, 3 July 2020[12]).

4.4. Human resources management field

LVA also has important applications in human resource management. For example, in the recruitment process, many interviewers have long since learned how to successfully answer provocative questions and even lie. LVA technology, on the other hand, claims to be able to detect the smallest variations in the characteristics of the interviewer's voice when answering particular questions, which are beyond the reach of human hearing. The LVA system shows how pronounced the fluctuations in the interviewer's voice are and assesses whether he is confident in his answers and how emotionally and mentally stressful they are. Based on this data, the company can determine the reliability of the information provided by the interviewer and make a hiring decision (HR in Russian, 1 November 2018[13]). However, it is also important to note that this aspect of the application needs to comply with the provisions of the relevant laws and regulations, to weigh the potential risks, and to avoid violating the privacy of the candidates (Santos, 2023).

4.5. Mental health assessment field

Some researchers have also explored the potential and application of LVA in preventing juvenile recidivism. They argued that LVA could be used to analyze voice samples to identify juveniles at risk for aggressive or violent behavior. This would allow for early, targeted interventions to address underlying issues that may lead to criminal activity, such as anger management problems, frustration, or social conflict (Chaube and Liberman, 2024).

Beyond the aforementioned fields, researchers have also explored the applicability of LVA technology in other domains. For instance, Rozenberg et al. from the Faculty of Aeronautics at the Technical University of Košice, Slovakia, investigated the scientific challenges associated with applying LVA technology to airport target protection processes (Rozenberg et al., 2019).

5. Case study

5.1. Successful cases

In the field of forensic investigation, India has been among the first to adopt LVA technology for case investigations across multiple regions. Since its introduction in 2012, this technology has been repeatedly deployed in criminal interrogations, including those involving high-value targets. By analyzing changes in tone, intonation, and frequency during their responses, the technology assists police in uncovering critical clues regarding their training background, criminal intent, and associated individuals, thereby breaching suspects' psychological defenses. In 2014, Mumbai police employed LVA technology to interrogate five suspects accused of participating in a gang rape case, prompting the

[8] JRvoice. 2 April 2019. http://www.jrvoice.cn/index.php?c=article&id=62

[9] OMMCOM NEWS. 4 February 2023. https://ommcomnews.com/odisha-news/naba-das-murder-case-accused-gopal-undergoes-lva-test/

[10] 10 NIKKEI. 6 October 2020. https://www.nikkei.com/article/DGXMZO64653230W0A001C2X30000/

[11] Customer Service Manager. 12 September 2019. https://www.customerservicemanager.com/nemesysco-emotion-analysis-technology-improves-japan-call-center-operations/

[12] ASIA INSURANCE REVIEW. 3 July 2020. https://www.asiainsurancereview.com/News/View-NewsLetter-Article/id/72457/type/eDaily/China-CPIC-reduces-life-insurance-fraud-with-voice-analytics-and-emotion-detection-solution , Nov. 29, 2024.

[13] HR in Russian. 1 November 2018. https://hr-elearning.ru/hr-budushhee-analiz-golosa-pomozhet-vybrat-kandidata/?fbclid=IwAR2YX5Az8Bw xJVxF9WQbimbesShhvsIAutEcNCEwu3SndSbeLKBWMM-z2GU

defendants to confess to two additional gang rape and molestation cases. In 2015, Mumbai's Criminal Investigation Department administered LVA tests to suspects in a murder case. By recording and mapping vocal fluctuations, they assessed the credibility of confessions, providing crucial direction for solving the case (Kacker and Shukla, 2020).

Rathod et al. initiated a forensic psychological analysis of recorded calls in a sex extortion case through Layered Voice Analysis (LVA) technology to validate the fraudster's deception. In this case, a man accepted a friend request from an unknown woman on social media and participated in an explicit video call, which was recorded by the fraudster for blackmail. Initially, the man paid a total of 35,000 Indian rupees in four installments. Subsequently, he was deceived by another fraudster posing as an official from the Central Bureau of Investigation (CBI). The scammers demanded an additional 27,500 Indian rupees to "delete the pornographic videos that had already been circulated," granting only 40 min to make the payment. Having already suffered losses, the man became more vigilant this time. He recorded the call, refused to pay again, and submitted the recording and chat screenshots to researchers. Researchers employed LVA technology to conduct offline analysis of the call recording, categorizing segments as follows: the man was labeled "Primary Subject," the fraudster as "Secondary Subject," noise and overlapping speech as "Mixed," and the most critical segments as "Relevant." The system color-codes the message through three levels (Level 1 contains emotional messages such as high anticipation and stress, Level 2 contains Intensive Emotional Reaction messages, and risk messages such as inaccuracy and medium risk) were generated. After reviewing all 168 segments, 31 stressed segments, 42 highly stressed segments, 37 inaccuracy segments, and 24 medium risk segments were detected from the fraudster's voice segments, and only 7 segments were truth, whose voice characteristics reflect a motivation to deceive (Rathod et al., 2021).

In a lie detection practice involving a murder case, LVA also demonstrated certain practical value. In this case, the victim was killed by his girlfriend using a toxic herbal mixture. The perpetrator, whose family had arranged a suitable marriage for her, sought to end the relationship with the victim. After being rejected, she meticulously planned the murder. Following the incident, the victim's brother recorded and released phone calls and WhatsApp voice messages with the perpetrator. These public recordings became the core data for the analysis. Analysis from two LVA sessions revealed the perpetrator exhibited elevated Emotional Level (Sub EMO Level), indicating fluctuating stress, anxiety, and deception; low Cognitive Level (Sub COG Level), suggesting her responses may have been rehearsed or influenced by external factors; and an abnormal Anticipation Level, implying premeditation or deceptive narrative control.

Only about a dozen percent of the segments of the killer's statement were categorized as true TRUTH, with nearly one hundred segments showing markers of INACCURACY, EXTREME STRESS, and HIGH TENSION. Key segments involving discussion of topics such as poisons, drugs, and marital status all triggered deception or stress markers. Overall, the killer's statements were characterized by clear information withholding and manipulation, and her voice profile supported the conclusion of "prior knowledge and deliberate planning of the murder". These results are corroborated by the physical evidence verified by the police and the court's judgment, which show that LVA can uncover the psychological motivation and truth behind the words in the lie detection of complex criminal cases, thus reducing the investigation time and bringing convenience to the related work (Savithri and Sylaja, 2025).

In addition to the cases mentioned above, multiple news reports in recent years indicated that LVA has been successfully used in several commercial entities. This technology has been instrumental in a number of areas, helping to solve problems encountered by some commercial entities. For example, on Saturday, January 29, 2011, at 17:41, a 4.3 magnitude earthquake struck Hungary, and the next day, insurance companies received 163 claims by phone. After analyzing them through LVA (the product is InTone.ai Risk), it was found that 119 of them were recognized as true claims and 44 as fraudulent claims (Nemesysco, 29 November 2024[14]). In 2014, a South African insurance company flagged 30,000 claims as risky, then used LVA (the product is InTone.ai Risk) to conduct brief interviews with claimants and determined that about 20,000 claims were "low risk" and would not be investigated further. The company believes that this not only improved customer service for legitimate claimants, but also saved approximately $15,000,000 in travel and expenses (based on an investigation cost of $750 per case) (Nemesysco, 29 November 2024[14]). Additionally, according to news reports, a suspect in India was arrested on charges of planning and carrying out a bombing. Since the suspect refused to undergo a conventional polygraph test, experts from the Central Forensic Science Laboratory in Delhi employed LVA equipment to administer a lie detection test. The test ultimately proved successful, with the suspect truthfully confessing to the criminal acts (THE TIMES OF INDIA, 1 February 2019[15]).

The cases above are sourced from the official website of the company that owns LVA technology. Although the company claims LVA technology has been successfully applied in insurance fraud detection, these cases lack independent peer review. Based solely on the available information, it isn't easy to gain recognition from the academic community. Furthermore, while the case of Indian experts successfully identifying a criminal suspect using LVA demonstrates the technology's potential in deception detection, it also presents uncertainties. Academic researchers cannot definitively determine whether this outcome was merely coincidental or influenced by the operator's subjective judgment.

5.2. Failure cases

Beginning in 2008, the Department for Work and Pensions (DWP) spent more than £2 million on a trial to evaluate the ability of LVA (known as Voice Risk Analysis, VRA, in the UK) to identify fraudsters in the benefit review process, providing important data on which to base judgments about the validity and reliability of the technology. From the results of the evaluation, the area under the curve (AUC) of the work characteristics of the subjects in the seven regions fluctuated between 0.51 and 0.73, and the results were shown to be due to coin flips in only three of the seven areas. One of the best performing results corresponded to a discrimination index (d') of about 0.9, a poor performance. Moreover, many cases flagged as "high-risk" by LVA technology were ultimately proven legitimate; after more detailed interviews, the benefits for the relevant applicants remained unchanged. Based on the research, it can be inferred that the statistical results presented in the assessment reports likely incorporate the subjective judgments of the certifying personnel at the company owning the LVA technology. These certifiers, while listening to the recorded interviews, may make uncontrolled corrections and adjustments to the meaningless raw data generated by the LVA technique, based on personal experience and non-standardized interpretation logic (Lacerda, 2009). After the DWP's trials ended, it was announced in 2010 that it had abandoned the use of the technology to catch a slice of welfare recipients. (The Guardian, 21 November 2025[16]).

Other examples of failures in the use of LVA can be found in some of the published research articles mentioned above. These articles document in detail the challenges and lessons learned from failures in the use of LVA in various scenarios. Examples include studies by Prof. Kelly Damphousse et al. and Prof. Frank Horvath et al. (Damphousse, 2008;

[14] Nemesysco. 29 November 2024. https://www.nemesysco.com/case-studies/

[15] THE TIMES OF INDIA. 1 February 2019. https://timesofindia.indiatimes.com/city/bhubaneswar/crime-branch-submits-chargesheet-in-parcel-blastcase/articleshow/67789035.cms?utm_campaign=andapp&utm_medium=referral&utm_source=native_share_tray

[16] The Guardian. 21 November 2025. https://www.theguardian.com/science/2010/nov/09/lie-detector-tests-benefit-cheats

Damphousse et al., 2007; Horvath et al., 2013). These failures show that LVA's ability to detect spoofing is clearly lacking, and there is a longer road to follow.

Discussions and lessons learned about LVA failures suggest that the effectiveness of the LVA system, as reported (i.e., by law enforcement and intelligence personnel), may actually be attributable to experienced interrogators keenly capturing clues directly from interview results rather than with the aid of the output of the LVA system. The samples of audio recordings analyzed by the LVA operators contained only the subjects' statements and reactions related to the events under investigation, leading Srivastava et al. to suggest that the poor performance of the LVA operators here was due to their lack of access to external information, which is crucial for the LVA operators (Srivastava et al., 2022).

6. Challenges for LVA

Although the company behind LVA technology provides a general overview of its analytical process on its official website, the vast majority of its internal procedures remain undisclosed. Consequently, we cannot make definitive judgments about the technology. Regarding the publicly available aspects, the scientific validity and rationality of LVA technology continue to be questioned by some linguists. In 2007, two researchers conducted an in-depth analysis of LVA, including the patented algorithm on which LVA is based. They concluded that its creative analysis techniques lacked credibility. First, the sampling frequency in signal processing is relatively low, which leads to the loss of speech details and the inability to capture "subtle changes," meaning it is too coarse to capture effective information. Second, the patent's linking of information about Thorn and Plateau to tones as a means of detecting an individual's mood was viewed with deep skepticism. This is because Thorn and Plateau are mainly obtained by comparing the amplitude of adjacent samples, which depends on both the particular articulation and the recording method; Third, the researchers also questioned LVA's claim to "detect brain activity traces through speech" because when the researchers searched for neuroscience and acoustic papers, they found no evidence that "speech signals contain brain activity traces" (Eriksson and Lacerda, 2007). Later, one of the researchers published two more articles in 2009 and 2013 criticizing LVA technology, focusing primarily on its shortcomings in signal processing and using real-world application data to demonstrate the "ineffectiveness" of LVA technology (Lacerda, 2009, 2013).

In addition, after calculating the relevant parameters, the LVA technology also provides the normal range for different parameters. For example, for the emotional level parameter, the normal range for males is between 100 and 300, while for females, it is between 200 and 400 (CHAO et al., 2020). Whether such boundaries are reasonable and whether they are influenced by factors such as cultural background and dialect are questions worth exploring. And as far as we know, the derivation of thresholds for different emotions needs to be tested and examined through experiments. Then, this experimental process faces a difficulty and a challenge in defining the thresholds of different emotions with the control of the purity of each emotion. Speech-based emotion detection follows the same logic as polygraphs in that the essence of both is to capture and analyze different physiological signals. These physiological activities are generated and driven by different neural activities. Many functional Magnetic Resonance Imaging (fMRI) studies have met some troubles in differentiating the basic emotions (such as fear, anger, joy, sad, and disgust) with distinct universal signals, physiology, especially the localization of the central nervous system (Lindquist et al., 2012; Lindquist and Barrett, 2012). This is because emotions are not based on a simple one-to-one mapping between neural structures and functions, but rather a more complex network-based one-to-many relationship between neural structures and functions (Celeghin et al., 2017). Therefore, emotions are generally present in a mixture, and the presence of a single emotion is extremely rare. Similar to the presentation of sound waves in nature, sound waves are generally not presented in the form of single waves; they are presented in the form of complex waves, and separating complex waves into single waves is a very difficult process. So how the LVA technology differentiates between emotions in experiments, and how it arrives at the models and algorithms it operates on, is unknown. In summary, LVA technology faces the same challenge as other emotion detection techniques: how to obtain ground truth data for emotions when constructing the LVA system. Acquiring ground truth remains a major hurdle in emotion detection, particularly when emotion categories are further subdivided. This significantly increases the difficulty of obtaining accurate ground truth, as emotions rarely manifest in isolation but are often a blend of multiple sentiments.

Specifically, only technologies that withstand rigorous testing and operate on scientifically sound principles can gain public acceptance. Although LVA has been successfully implemented in certain commercial entities (such as insurance companies and recruitment centers), critical institutions (such as judicial bodies) must exercise caution when considering LVA assessments as decision-making criteria. This is because erroneous accusations based on LVA results could lead to severe consequences.

Finally, because LVA is a covert testing technique, it can be used to test speakers without their knowledge. It is important to think carefully about the range of ethical issues that this may raise and to assess the various possible consequences to ensure that the decisions made are both ethical and capable of contributing to the harmonious development of society.

7. The future of LVA

A great advantage of LVA is that it has both on-line and off-line modes, and its on-line mode is virtually automatic and does not require much from the operator. However, studies have shown that its online mode is less sensitive than its offline mode (KS et al., 2024). In addition, LVA will likely play an important role as a mood detection tool in a number of domains. For example, LVA can be used in call centers to help staff understand the emotions expressed by customers in real time and provide better service; it can also be used in banks to help them determine the integrity of loan users; it can also be used in the insurance industry to help insurance companies determine whether there are fraudulent claims; LVA can also help company managers in the recruitment or selection of personnel, or even monitor the stress situation of employees to prevent brain drain, etc. (Sean Mitchell, 31 May 2022[17]). However, a rigorous and scientific technical review should be conducted to assess the effectiveness of LVA in detecting emotions or lies before applying it to multiple domains. To date, many of the emotion indicators generated by LVA technology have not been validated by peer review.

The advent of artificial intelligence has rendered the integration of AI into a multitude of domains an inevitable phenomenon. Consequently, the integration of AI methodologies with LVA has the potential to enhance the capacity to discern emotional states. Moreover, LVA is concerned with the analysis of emotional states expressed through speech, while the content of the speech itself is not a primary focus. This approach may result in an inaccurate interpretation of the communicator's intent, given the close relationship between the specific expression of speech and the conveyance of emotion. The combination of speech meaning and emotion analysis may result in a notable enhancement of the accuracy of the analysis, thereby providing more reliable results in a number of application scenarios.

[17] Sean Mitchell. 31 May, 2022. https://itbrief.asia/story/q-a-behavioural-cues-discuss-what-s-next-for-voice-analytics

8. Conclusion

LVA technology claims to be a method that detects emotional states, stress levels, and potential deception through psychoacoustic and physiological characteristics in voice signals. The focus is not on the words themselves, but rather on the state information that is latent behind them. LVA utilizes a distinctive mathematical process to discern and quantify the various forms of stress, cognition, and emotion manifested in the nuanced characteristics of the voice (Sean Mitchell, 31 May 202,2[15]). Although the technique has shown some potential for application in a number of areas, its accuracy and reliability are still controversial in both academic and practical applications.

For the time being, the academic community holds mixed opinions on the effectiveness of LVA (Benson et al., 2024; Eriksson, 2006; Feldman, 2016; Jain et al., 2024; Pietruszka, 2009; Scheerschmidt and Metzler, 2024; Verschuere, 2025). Due to the confidentiality of LVA technology, the specific algorithms used to calculate emotional features based on speech parameters have not been fully disclosed, resulting in limited academic transparency. Experts in the academic field are unable to fundamentally verify the overall effectiveness and reliability of the technology. Therefore, they can only make judgments based on experimental results and real-world application outcomes.

In the future, LVA still needs to break through academic controversies and industry resistance, and the marketization process of LVA relies on continuous technical verification. In the long run, if we can further improve the accuracy of the algorithm, accumulate more empirical data in the scene, and, along with the deepening of the research related to emotion and behavior, its application potential is expected to be further explored. However, whether its future development prospects are favorable still depends on the development of technical principles and industry acceptance.

Overall, despite some academic debate, LVA remains the primary tool currently capable of achieving non-contact, long-distance covert measurement. Our research analyzes LVA technology from an academic perspective. Just as the scientific basis of polygraph testing itself remains controversial to this day—particularly regarding comparative problem testing paradigms—this does not diminish its recognition and application in practical fields. We must remember that when deploying a new technology in practical applications, its effectiveness requires rigorous scrutiny to ensure it is a scientifically sound and reasonable tool.

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CRediT authorship contribution statement

Noé Xiu: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Wenmei Li:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Yuan Wang:** Writing – original draft, Investigation. **Béatrice Vaxelaire:** Writing – review & editing. **Rudolph Sock:** Writing – review & editing. **Fabrice Marsac:** Writing – review & editing. **Zhenhua Ling:** Writing – review & editing.

Declaration of competing interest

All authors declare no real or potentially perceived conflict of interest, including financial, personal, or other relationships with other organizations or companies that may inappropriately impact or influence the research and interpretation of the findings.

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Data availability

No data was used for the research described in the article.

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